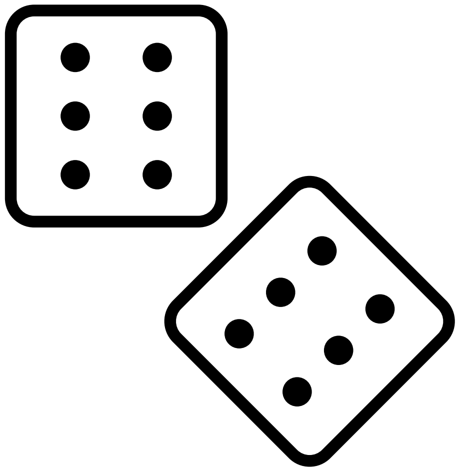
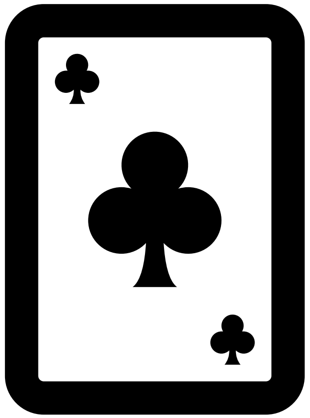




Probability

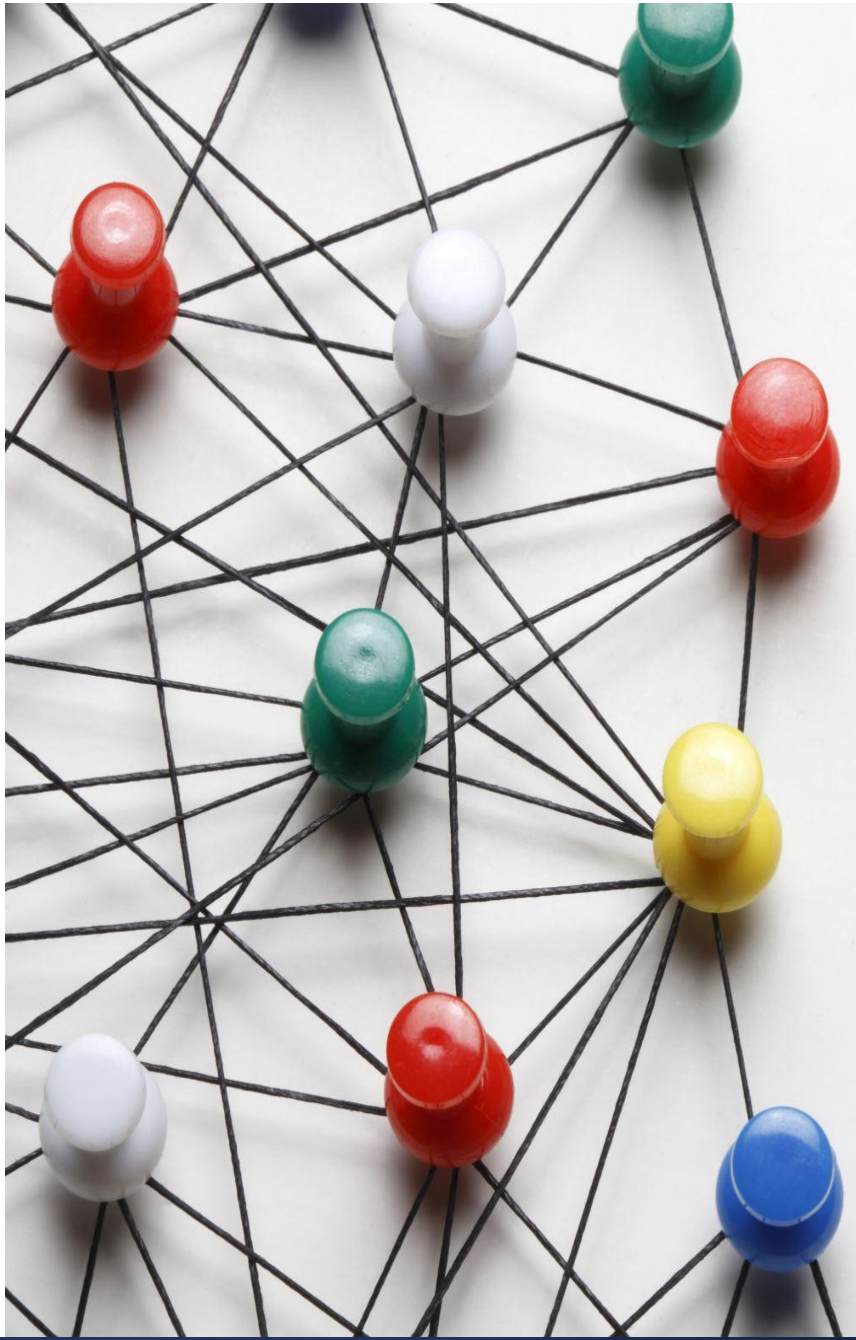
Probability is simply how likely something is to happen

A fun riddle for you

What has a head and a tail but no body?

Whenever we are unsure about the outcome of an event. We can talk about the probabilities of certain outcomes- how likely they are

Example: if you are flipping a fair coin, how can you quantify which one will you get, tail or head in such scenario we make use of probability



Why probability is important?

- **Accurate Predictions:** Probability helps scientists to make accurate predictions by quantifying the likelihood of different outcomes.

For example, in weather forecasting, meteorologists use probability models to predict the chance of rain, enabling people to plan their activities accordingly

- **Understanding Uncertainty:** Probability helps in understanding and managing uncertainty in data.

For instance, in stock market analysis, data scientists use probability models to estimate the probability of a stock's price going up or down, enabling investors to make informed decisions.

- **Modeling Uncertain Events:** Probability allows data scientists to model and analyze uncertain events.

For example, in insurance risk assessment, probability models are used to estimate the likelihood of certain events, such as car accidents or property damage, helping insurance companies determine premium rates.

- **Decision Making:** Probability helps in making optimal decisions based on available data.

For instance, in medical diagnosis, probability models can be used to calculate the probability of a patient having a particular disease based on symptoms and test results, aiding doctors in making accurate diagnoses.

Probability is important …

- **Framework for Other Concepts:** Probability provides a foundation for various other concepts in data science. For example, machine learning algorithms often use probability theory to estimate the likelihood of different outcomes and make predictions.

Usage of probability: probability is used in various domain. Some of them are

- Data Science
- Insurance
- Genetic Science
- Clinical trials
- Vaccination efficacy testing

Usage of probability in Statistical analysis and Data Science

- **Confidence Intervals**

It helps to know the probability of our data lying within a given interval.

- **Probability distribution**

Most of the data follows a certain distribution, some follow a normal distribution, Poisson distribution, Bernoulli distribution, Binomial distribution, while some others follow Exponential distribution

- **Bayes Theorem AND Naive Bayes Algorithm**

Bayes Theorem is used in machine learning, specifically in the Naive Bayes algorithm. This algorithm is commonly used for text classification, spam filtering, and sentiment analysis. It calculates the probability of a certain event occurring given the prior knowledge of related events.

- **Conditional Probability**

Conditional probability is used in various Statistical and data science applications, such as recommendation systems. It helps determine the likelihood of a certain event occurring given the occurrence of another event. For example, in a movie recommendation system, conditional probability can be used to predict the probability of a user liking a certain movie based on their previous ratings and preferences.

Usage of probability in Statistical analysis and Data Science Cont….

- **Central Limit Theorem**

The Central Limit Theorem is a fundamental concept in statistics and is widely used in Statistical analysis and data science. It will be discussed in detail in the next lecture. This theorem allows scientists to make inferences about a population based on a sample, enabling hypothesis testing and confidence interval estimations.

- **Markov Chains**

Markov Chains are used in a variety of Machine Learning applications, such as natural language processing and speech recognition.

<u>**These concepts play crucial roles in various techniques and algorithms, enabling scientists to make predictions, classify data, analyze patterns, and gain insights from complex datasets.**</u>

Key Terminology of Probability Theory

Random experiment: It is an experiment in which outcome is not known with certain

Sample space: It is a universal set that consists of all possible outcomes of an experiment. An example: outcome of a college application S = {admitted, not admitted}

Event: It is a subset of sample space and probability is usually calculated with respect to an event.
Example: Chances of getting head on a fair coin.

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Probability and Rules of Probability

In General, the probability score is given by the number of outcomes divided by the total number of outcomes in the sample space

$$P(A) = \frac{n}{N} = \frac{Number\ of\ outcomes\ A}{Total\ number\ of\ oucomes\ in\ sample\ space}$$

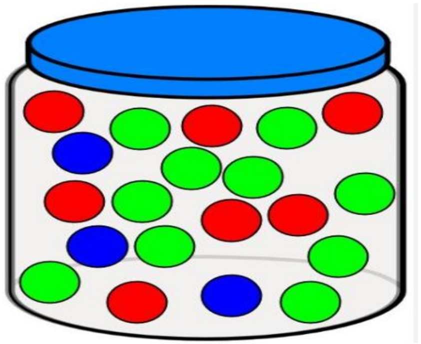
Rules of probability

Rule I:

The probability of an impossible event is zero. The probability of a certain event is one, Therefore, for any event A, the range of possible probability is

$$0 \leq P(A) \leq 1$$

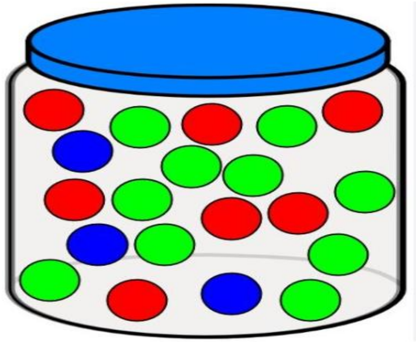
Example: Calculate the probability of drawing yellow ball?



Answer: P(yellow) = 0

Rule II:

For S the sample space of all possibilities, P(S) = 1. That is the sum of all the possibilities for all possible evens is equal to one.



Example: Calculate P(Green), P(Red), P(Blue), and P(S)

P(Green) = 10/20=0.5

P(Red) = 7/20=0.35

P(Blue) = 3/20=0.15

P(S) = P(Green)+ P(Red)+ P(Blue)= 0.15+0.35+0.5=1

Rule III:

For any event A, $P(A^C) = 1 - P(A)$. It follows then that $P(A) = 1 - P(A^C)$, where A^C is the complement of the event A.

Example:

The probability of not have red $P(Red^C)$

$P(Red^C) = 1\text{-}P(Red) = 1 - 0.35 = 0.65$

Marginal Probability and Joint Probability

In order to understand let us have an example from Netflix (a survey). It is a survey for 500 answers

Frequency distribution table

	Male	Female	Total
Breaking Bad	100	25	125
Money Heist	80	120	200
Other	50	125	175
	230	270	500

To convert it to **probability distribution table**, simple you will divide all entries by 500.

	Male	Female	Total
Breaking Bad	0.2	0.05	0.25
Money Heist	0.15	0.24	0.4
Other	0.1	0.25	0.35
	0.46	0.54	1

Joint probability is a statistical measure that calculates the likelihood of two events occurring together at the same time

Marginal probability is the probability of an event irrespective of the outcome of another variable

	Male	Female	Total
Breaking Bad	0.2	0.05	0.25
Money Heist	0.15	0.24	0.4
Other	0.1	0.25	0.35
	0.46	0.54	1

Question 1: What is the probability of a Netflix subscriber being Male?

Answer: 0.46

Question 2: What is the probability of a Netflix subscriber preferring Money Heist?

Answer: 0.4

Question 3: What is the probability of a Netflix subscriber preferring Breaking Bad and being Male?

Answer: 0.2

It is referred as the intersection between the two events $\boldsymbol{P(A \cap B)}$

Question 4: What is the probability of a Netflix subscriber preferring Breaking Bad or being Female?

	Male	Female	Total
Breaking Bad	**0.2**	**0.05**	**0.25**
Money Heist	**0.15**	**0.24**	**0.4**
Other	**0.1**	**0.25**	**0.35**
	0.46	**0.54**	**1**

Answer: 0.2+0.05+0.24+0.25 = 0.74

It is referred as the union between the two events $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Let us use the above formula by assuming A(Breaking Bad) and B (Female), this means

$$P(A \cup B) = 0.25 + 0.54 - 0.05 = 0.74$$

Question 5: Nur is a new Netflix subscriber, what is the chance that he would like breaking bad?

Answer: As you can see Nur is a male and we basically need to calculate the probability of liking breaking Bad given that he is a male. This scenario is called Conditional probability.

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

It is the probability of event A given B has occurred equal to the intersection of probability A and B (Both of them occurred together) divide by the probability of event B.

$$P(Breaking\ |Male) = \frac{0.2}{0.46} = 0.43$$

Question 6: Marie is anew Netflix subscriber, what is the chance that she would like Money heist?

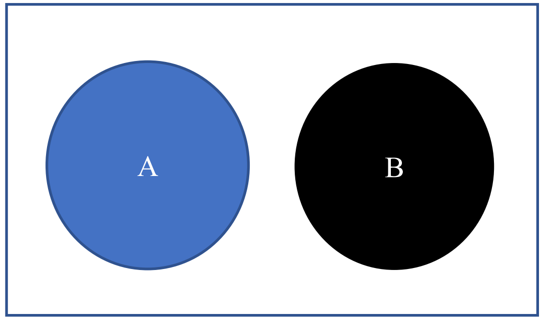
Answer: it is a conditional probability

$$P(Money|Female) = \frac{0.24}{0.54} = 0.44$$

Disjoint and Non-Disjoint Events

Disjoint Events are the events that cannot happen at the same time. It is also known as Mutually exclusive events.

Written in probability notation, events A and B are disjoint if their intersection is zero. This can be written as:



P(A and B) = 0

$P(A \cap B) = 0$

Examples:

- The outcome of a single coin toss can not be a head and tail at the same time.
- A student can not both fail and pass a same subject

Union of Disjoint Events

We know that $P(A \cup B) = P(A) + P(B) - P(A \cap B)$, since the events A and B are disjoint, this means that their intersection is zero.

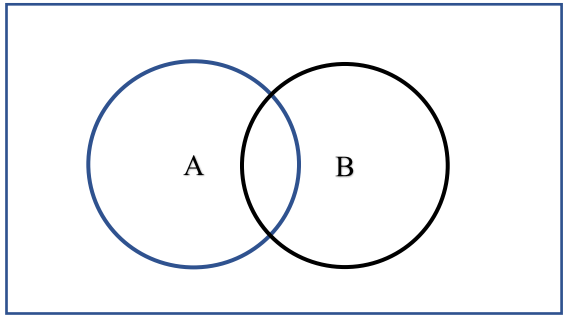
The union of disjoint event $P(A\ or\ B) = P(A \cup B) = P(A) + P(B)$.

Non-Disjoint Events, where events can happen at the same time.

Example: A student can get an A in statistics and A in History at the same time. This means

$$P(A \text{ and } B) \neq 0$$

$$P(A \cap B) \neq 0$$



Union of Non- Disjoint Events

$$P(A \text{ or } B) = P(A \cup B) = P(A) + P(B) - P(A \cap B) .$$

Dependent and Independent Events

Dependent events are those events that are affected by the outcomes of other events. i.e. Two or more events that depend on one another are known as dependent events. If one event is changed, then another is going to change.

Examples:

- Getting into a traffic accident is dependent upon driving or riding in a vehicle
- If you park your car illegally, you are more likely to get a parking ticket
- You must buy a lottery ticket to have a chance at winning, the likelihood of winning are increased if you buy more than one ticket.

Independent Events are those events whose occurrence is not dependent on any other event. If the probability of occurrence of an event A is not affected by the occurrence of another event B, then A and B are said to be independent events.

Example, the color of your hair has absolutely no effect on where you work. The two events of "having black hair" and "Working in a google" are completely independent of one another.

- Taking an uber and get a free meal at your favorite restaurant.
- Getting 6 on a dice and getting a promotion at work.

Product Rule for Dependent and Independent Events

The multiplicative rule of probability or product rule is used to find the probability of an intersection of events or two events happening at the same time.

There are two forms of this rule

- Specific Multiplication Rule
- General Multiplication Rule

For Specific Multiplication Rule

- It is used to calculate the joint probability of independent events.

$$P(A \cap B) = P(A) \times P(B)$$

Remember that in the independent events, the occurrence of event A does not affect the likelihood of event B.

Example 1:

Calculate the probability of obtaining “head” during two consecutive coin flips

Answer: In the first flip, P(H)= 0.5 and P(H) = 0.5 in the second flip. The probability of getting head in the first flip and getting head in the second flip are P(Head ∩ Head)= 0.5x0.5 = 0.25

Example 2:

You have 10 pairs of pants, out of them 3 are blue and you have got 16 shirts, out of them 4 are red. Selecting pants doesn't affect the likelihood of drawing the red shirt.

Answer: the probability of getting blue pants is P(Blue_pant) = 3/10 = 0.3 and the probability of getting red shirt P(red_shirt) = 4/16 = 0.25. The probability of getting blue pants and red shirt is P(red_shirt ∩ Blue_pants)= 0.25x0.3 = 0.075

<u>In these situations, we apply the specific multiplication rule</u>

General Multiplication Rule

This rule is used to calculate the joint probability for either independent or dependent events.

$$P(A \cap B) = P(A) \times P(B|A)$$

The joint probability of A and B occurring equals the probability of A occurring multiplied by the conditional probability of B occurring given that A occurred.

Remember that

$$\boldsymbol{P(B|A) = \frac{P(A \cap B)}{P(A)}}$$

In the case of independent events

$$\boldsymbol{P(B|A) = P(B)}$$

Because the occurrence of B has no effect on the occurrence of A.

Example 1: Nur has to select two students from a class of 23 girls and 25 boys. What is the probability that both students chosen are boys?

Answer:

Total number of students = 23 + 25 = 48

The probability of choosing the first boy, say B1 = 25/48

The probability of choosing the second boy, say B2 = 24/47

Then,

$$P(B1 \text{ and } B2) = P(B1) \times P(B2|B1)$$

$$= (25/48) \times (24/47) = 0.265$$

Example 2: A bag contains 6 red, 5 blue, and 4 yellow balls. 2 balls are drawn, but the first ball is drawn without replacement.

Find the following:

a) P (red, then blue)

b) P (blue, then blue)

Answer a):

There are six red balls and a total of fifteen balls.

$$P\ (\text{red}) = 6 / 15$$

The probability of the second draw affected the first.

Number of blue balls = 5

Total number of balls left = 14

$$P\ (\text{drawing blue after red}) = 5 / 14$$

$$P(\text{drawing red, then blue}) = P\ (\text{drawing red}) \times P\ (\text{blue after red}) = (6/15) \times (5/14) = 0.143$$

Answer b):

Number of blue balls = 5

Total number of balls left = 15

The probability of drawing a blue ball = 5/15

The probability of the second draw affected the first.

Now, there are 4 blue balls left and 14 balls left.

P (drawing a blue ball after a blue ball) = 4/14

P (blue, then blue) = P (drawing blue ball) x (drawing a blue ball after a blue ball)= (5/15) x (4/14)

Hence, the probability of drawing a red ball followed by a blue ball is 2/21.

Example 3: A wallet contains 4 bills of 5 dollars, 5 bills of 10 dollars, and 3 bills of 20 dollars. 2 bills are chosen randomly without replacement. Find P (drawing a 5-dollar bill followed by a 5-dollar bill).

Answer:

P (drawing a 5-dollar bill) ?

Number of 5-dollar bills = 4

Total number of bills = 12

P (drawing a 5-dollar bill) = 4 / 12

P (drawing a 5-dollar bill after a 5-dollar bill)?

The probability of the second draw affecting the first.

Number of 5-dollar bills left = 3

A total of 11 bills are left.

P (drawing a 5-dollar bill after a 5-dollar bill) = 3/11

P (drawing a 5-dollar bill followed by a 5-dollar bill) = P (drawing a 5-dollar bill) * P (drawing a 5-dollar bill after a 5-dollar bill) = 4/12 x 3/11 = 0.91

Hence, P (drawing a 5-dollar bill followed by a 5-dollar bill) = 0.91.

Example 4: A bag contains 4 red, 3 pink and 6 green balls. Two balls are drawn, but the first ball drawn is not replaced.

a) Find P(red, then pink)

b) Find P(pink, then pink)

Answer a):

Number of red balls = 4

Total number of balls = 13

P(red) = 4/13

The result of the first draw will affect the probability of the second draw.

There are 3 pink balls.

Number of balls left = 12

P(pink after red) = 3/12 = ¼

P(red, then pink) = P(red) x P(pink after red) = (4/13) × (1/4) = 1/13

Hence, the probability of drawing a red ball and then a pink ball is 1/13.

Answer b):

Number of pink balls = 3

Total number of balls = 13

$$P(\text{pink}) = 3/13$$

The result of the first draw will affect the probability of the second draw.

Number of pink balls left = 2

Total number of balls left = 12

There are a total of 12 balls left.

$$P(\text{pink after pink}) = 2/12 = 1/6$$

$$P(\text{pink, then pink}) = P(\text{pink}) \cdot P(\text{pink after pink}) = (3/13) \times (1/6) = 3/78 = 1/26$$

Hence, the probability of drawing a pink ball and then a pink ball is 1/26.

The Law of Total Probability

- The Law of Total Probability is a fundamental concept in probability theory that allows us to calculate the probability of an event by considering all possible ways or scenarios that can lead to that event.
- It is often used when the event of interest can occur under different conditions or circumstances.
- Formally, let's consider an event B and a set of mutually exclusive events (disjoint events, it means that only one of them can occur at a time) and exhaustive events (it means that together they cover all possible outcomes or scenarios) $\{A_1, A_2, ..., An\}$.

The Law of Total Probability states that the probability of event B can be calculated as the sum of the probabilities of B occurring given each possible condition Ai, weighted by the probability of each condition Ai occurring. Mathematically, it can be expressed as follows:

$$P(B) = P(A_1) \times P(B|A_1) + P(A_2) \times P(B|A_2) + ... + P(An) \times P(B|An)$$

Example:

A company manufactures two types of smartphones: Type A and Type B. The company produces 60% Type A smartphones and 40% Type B smartphones. The defect rate for Type A smartphones is 5%, while the defect rate for Type B smartphones is 3%. Suppose that a customer buys a smartphone from this company, and we want to **calculate the probability that the purchased smartphone is defective.**

Answer:

A: The smartphone is of Type A.

B: The smartphone is defective.

We are given:

$$P(A) = 0.6 \text{ (probability of Type A smartphone)}$$

$$P(B|A) = 0.05 \text{ (probability of defect given Type A smartphone)}$$

$$P(\text{not } A) = 0.4 \text{ (probability of Type B smartphone)}$$

$$P(B|\text{not } A) = 0.03 \text{ (probability of defect given Type B smartphone)}$$

We want to calculate P(B), the probability that the purchased smartphone is defective.

According to the Law of Total Probability:

$$P(B) = P(A) \times P(B|A) + P(\text{not } A) \times P(B|\text{not } A)$$

Substituting the given values:

$P(B) = (0.6 \text{ x } 0.05) + (0.4 \text{ x } 0.03) = 0.03 + 0.012 = 0.042$

Therefore, the probability that the purchased smartphone is defective is 0.042 or 4.2%.

Bayes Theorem

It is mathematical formula for determining conditional probability. It shows the relation between a conditional probability and its reverse form.

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

<u>To prove it</u>

$P(A|B) = \frac{P(A \cap B)}{P(B)}$ and $P(B|A) = \frac{P(B \cap A)}{P(A)}$

Since $P(A \cap B)$and $P(B \cap A)$ are the same, then

$$P(A|B)\,P(B) = P(B|A)\,P(A)$$

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Examples:

- Given that test was positive, what is the probability of having a disease.
- Given that a person likes action, what is the probability that he would like to watch Kung Fu Panda.

Example:

In a certain game, the probability that a player is laying = 0.75 and the probability that a player is not laying = 0.25. The probability that a player wins when he lied = 0.43, the probability that a player wins when he didn't lie = 0.57. **Find the probability that a player was lying given that he won the game?**

Answer:

A: Player is lying

B: Player wins the game

We are given:

P(A) = 0.75 (probability that a player is lying)

P(not A) = 0.25 (probability that a player is not lying)

P(B|A) = 0.43 (probability of winning when lying)

P(B|not A) = 0.57 (probability of winning when not lying)

We need to find P(A|B), the probability that a player was lying given that h he/she won the game.

According to Bayes' theorem:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

To calculate P(B), we can use the law of total probability:

$$P(B) = P(A) \times P(B|A) + P(\text{not } A) \times P(B|\text{not } A)$$

Substituting the given values:

$$P(B) = 0.75 \times 0.43 + 0.25 \times 0.57$$

$$= 0.3225 + 0.1425$$

$$= 0.465$$

Now we can calculate P(A|B):

$$P(A|B) = (P(A) \times P(B|A)) / P(B)$$

$$= (0.75 \times 0.43) / 0.465 = 0.691$$

Therefore, the probability that a player was lying given that they won the game is approximately 0.691, or 69.1%.

Application of Bayes Theorem in Data Science

1. Hypothesis test

Hypothesis approximates a target function that needs to be tested on data. For example, the mean weight of a newborn baby is 3.5 kg.

Bayes theorem allows us to test whether the hypothesis holds true for a given data as, P(h|D) which means, the probability of the "hypothesis" being true for a given "data".

2. Classification

When the possible values are categorical. Bayes theorem can be applied to classification problems.

For example: whether a customer defaults on credit card payment or not based on the account balance.

- Naïve Bayes classifier is one implementation of Bayes theorem.

3. Model Optimization

Optimizing machine learning models involves finding an input that minimizes or maximizes an objective function. For example, if my objective function is related to accuracy, the objective is to maximize. If it is related to a cost may be the objective is to minimize.

- Bayes' theorem applied probability to find out these values,
- Bayesian optimization is a technique used to improve the performance of a machine learning model.

Random Variable

In the previous topics in this chapter, our explanation was based on coins, dice and other random experiment with a few outcomes

In reality,

Our main treatment

- In web, …. "subscriber, clicks, viewers ….
- Sales, …."Yield, weight, sales….
- Traffic, …."time, congestion, delay…..
- Medicine, …."age, temperature, heart rate…
- Student,…… "GPA, Tuition, assignment…

What are the common things in the above mentioned?

"Numbers"

Random variable is a real-valued function, defined over the sample space of a random experiment.

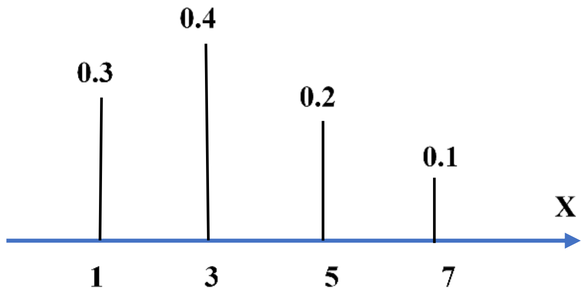
- It is a rule that assigns a numerical value to each outcome in a sample space.

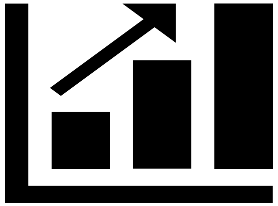
- Generally, the random variable is denoted with capital letters such as X and Y and defines the possible outcome values of an unexpected phenomenon

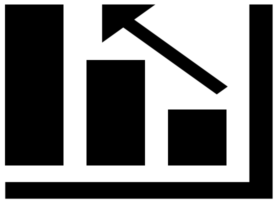
Why Random Variable ?

<u>With numerical values (numbers), the distribution of P(X) can be</u>

- View on a line
- Express as a function $P(X) = \frac{1}{2X}$
- Easy to consider the properties such as increasing or decreasing







Random variable X helps to

- Perform operation such as addition (X+1) and squaring (X^2)
- Combine variable X+Y
- Consider properties like the average value of X

Mainly, we have two types of Random vatable.

Types of Random Variable

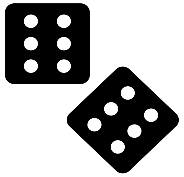
- Discrete random variable

- Continues Random variable.

In Discrete random variable,

- It is a variable that takes only a finite number of distinct values such as 0, 1, 2 and so on. Examples: {1,2,3}, {e, pi} or countably* infinite N (natural number) or Z (integer number)

Example: If we are interested in the number of heads in 3 fair coins

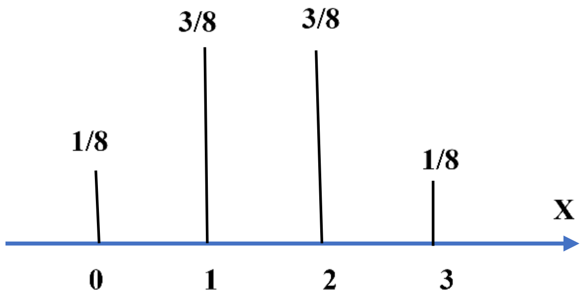


Ω = {ttt, tth, tht, thh, htt, hth, hht, hh}……| Ω |= 8

Each event gets same probability …P = 1/8

Let us denote X is the number of heads

X	Outcomes	P(X)
0	ttt	1/8
1	tth, tht, htt	3/8
2	thh, hth, hht	3/8
3	hhh	1/8



"Countable" means the values can be listed one by one, even if they are infinite.

Example 2: In four-sided dominoes, we got the following table

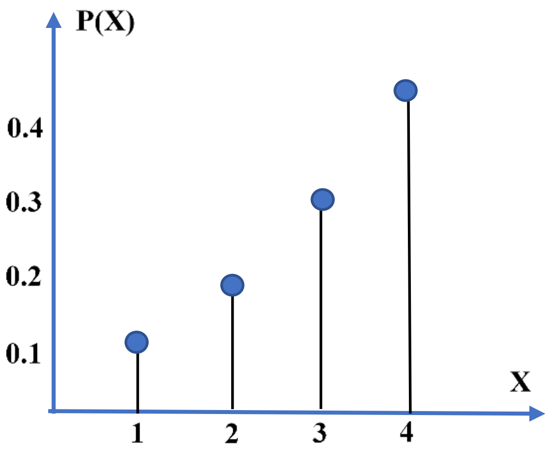
X	1	2	3	4
P(X)	0.1	0.2	0.3	0.4

It is explicit that P(1) = 0.1, P(2) = 0.2, P(3) = 0.3, P(4) = 0.4

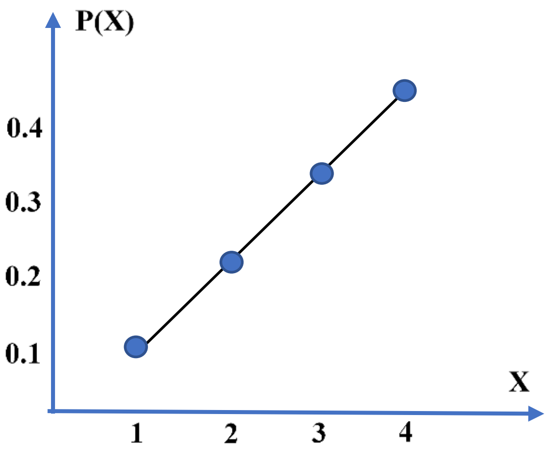
- Without number, you can make it as function $P(X) = \frac{X}{10}, \quad X \in \{1, 2, 3, 4\}$

- It can be presented by Graphs

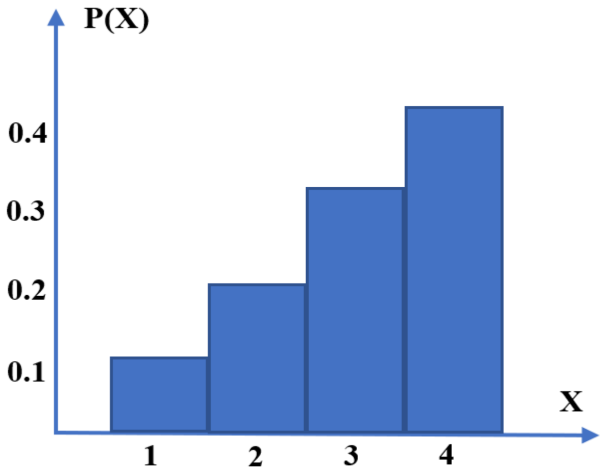
Stem plot



Line plot



Histogram



In Continues random variable

It is a variable that can take on any value of a specified domain (i.e., any value in an interval), e.g., uncountably infinite [0, 2], (-1, 4) U [4, 5), or R

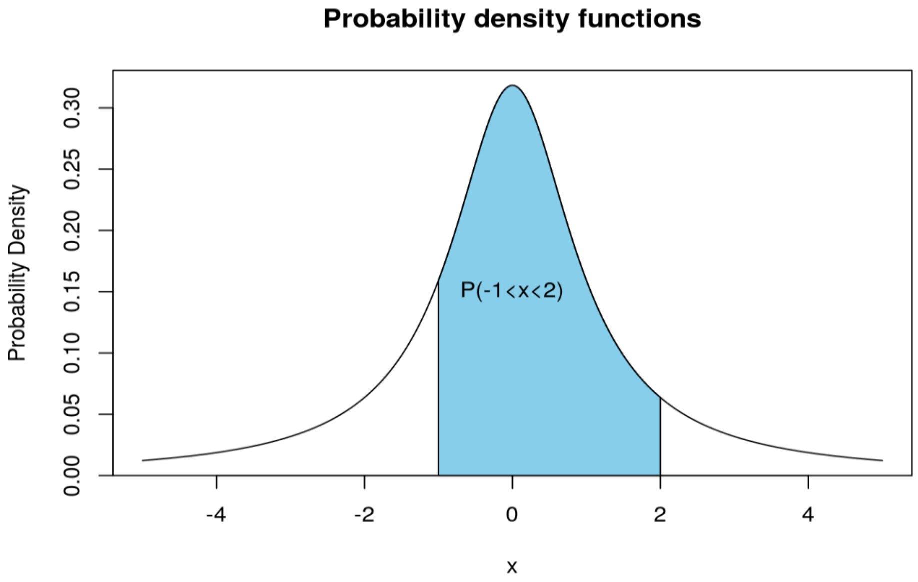
For example, the height of students in a class, the amount of tea in a glass, the change in temperature throughout a day, and the number of hours a person works in a week all contain a range of values in an interval, thus continuous random variables.

- The main difference between continuous and discrete random variables is that continuous probability is measured over intervals, while discrete probability is calculated on exact points.

- For example, it would make no sense to find the probability it took exactly 32 minutes to finish an exam. It might take you 32.012342472… minutes. Probability of points no longer makes sense when we move from discrete to continuous random variables.

- Instead, you could find the probability of taking at least 32 minutes for the exam

or the probability of taking between 31 and 33 minutes to complete the exam. Probability function distributions will help us in this regard. These functions help to determine the probability by finding the area under the function where the total area under the curve must equal 1.



These are different ways of specifying the distribution of a random variable

(Functions that describe the probability distributions of Random Variables)

Probability Mass Function (PMF)

- It is a function that describes the likelihood for a random variable to take on a given value. It is a mapping. P maps the sample space to real numbers $P : \Omega \rightarrow R$

- It is a way to describe the probability distribution of a discrete random variable.

- It is non-negative and its integral over entire sample space is equal to 1 or they sum to one $\sum_{X \in \Omega} P(X) = 1$

- It is not negative $P(X) \geq 0 \; where \; X \in \Omega$

- An example of PMF is Binomial Distribution

When we talk about random variables, sometimes we are interested in specific values like the probability that we will get exactly A+ in the class or the temperature is going to be exactly 27 degrees Celsius. More often we are more interested not in a particular value but rather in a range of values or in an interval of values.

- Temperature between 25 to 30 Celsius
- Salary > 80 k $
- GPA > 3.0

Probability Density Function (PDF)

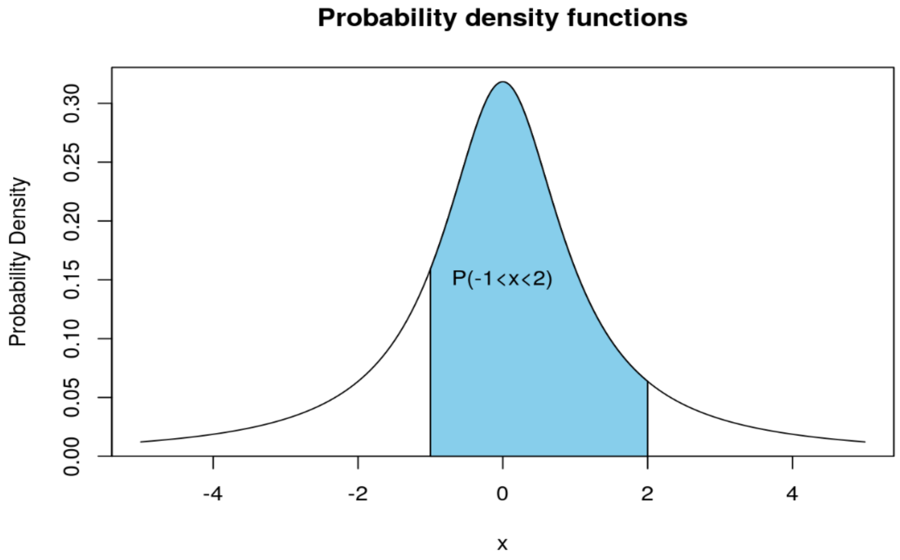
The probability density function produces the likelihood of values of the continuous random variable.

- It is defined as an integral of the density function of the random variable over a given range. It is denoted by f (x).
  - This function is positive or non-negative at any point of the graph, $f(x) \geq 0$
  - The integration of this function, more specifically the definite integral of PDF over the entire space is always equal to one.

$$\int_{-\infty}^{\infty} f(x)dx = 1$$

- If we find P(X = x), it does not work. Instead of this, we must calculate the probability o y function formula is given as,

$$P(a \leq X \leq b) = \int_a^b f(x)dx$$



- An example of PDF is a normal distribution

In short, PDF (probability density function) is defined for continuous random variables, whereas PMF (probability mass function) is defined for discrete random variables.

Cumulative Distribution Function (CDF)

It is a simple function that helps to determine interval probabilities called cumulative distribution function (CDF)

$$F(x) = P(X \leq x) = \sum_{u \leq x} P(u)$$

- It is a function that describes the probability that a random variable will take on a value less than or equal to a certain value.
- **It is a way to describe the discrete or continuous probability distribution of a random variable?**
- An example of CDF is Exponential distribution

In other words, a PDF gives the probability of a random variable being in a certain range, while a CDF gives the probability of a random variable being less than or equal to a certain value.

Properties of CDF

- Non decreasing

It means that the values of the CDF increase or stay the same as the input variable increases. In other words, as the input variable moves along the range of possible values, the CDF either remains constant or increases.

Mathematically, for any two values x_1 and x_2 in the domain of the random variable, if $x_1 \leq x_2$, then the corresponding CDF values $F(x_1)$ and $F(x_2)$ satisfy $F(x_1) \leq F(x_2)$.

- Limits: $\lim_{x \to \infty} F(x) = 1$ & $\lim_{x \to -\infty} F(x) = 0$

- Right continuous but not necessarily left continuous $\lim_{x \to a} F(x) = F(a)$

Question: Why is CDF and PDF useful?

Because it allows us to calculate the probability of intervals easily

We mentioned that CDF can be used for both discrete and continues random variables.

How does CDF work with Discrete random variables?

Let us elaborate on this question with an example.

Example: A professor wants to analyze the performance of their students in a class. The final exam scores are discrete and range from 0 to 100. The professor wants to determine the probability that a student scored less than or equal to a specific grade. The final exam scores follow a discrete distribution. The professor has the following data on the number of students who achieved each score:

Score	Number of Students
60	5
70	10
80	15
90	7
100	3

Solution:

To calculate the CDF, we need to sum up the probabilities of achieving a score less than or equal to a specific value.

$$CDF(x) = P(X \leq x)$$

For example, to find the probability that a student scored less than or equal to 80, we calculate the CDF as follows:

$$CDF(80) = P(X \leq 80) = P(X = 60) + P(X = 70) + P(X = 80)$$

To calculate each probability, we divide the number of students who achieved that score by the total number of students:

$$P(X = 60) = 5 / (5 + 10 + 15 + 7 + 3)$$

$$P(X = 70) = 10 / (5 + 10 + 15 + 7 + 3)$$

$$P(X = 80) = 15 / (5 + 10 + 15 + 7 + 3)$$

Then, we sum up these probabilities:

$$CDF(80) = (5 / 40) + (10 / 40) + (15 / 40) = 0.75$$

Therefore, the CDF at 80 is 0.75, indicating that there is a 75% probability that a student scored 80 or less on the final exam.

By calculating the CDF for different values, the professor can determine the probabilities associated with specific score ranges and make informed decisions about grading or assessing the performance of their students

Probability Distribution for Discrete Random variables

Binomial Distribution

- It is a discrete probability distribution of a random variable that has two outcomes, either Success or Failure.
- The objective is to find the probability of getting x successes out of n-trials

For example, if we toss a coin, there could be only two possible outcomes: heads or tails, and if any test is taken, then there could be only two results: pass or fail.

Note : We call a random experiment with two possible outcomes: success or failure as **Bernoulli trial**

- Probability of success is p, hence probability of failure is (1-p)
- It makes use of PMF and CDF

Binomial Distribution formula

$$Bi(X = x) = n_{C_x} P^x (1 - P)^{n-x}$$

where, n = the number of experiments, p = Probability of Success in a single experiment

and $n_{C_x} = \frac{n!}{x!(n-x)!}$

Example 1: If a coin is tossed 5 times, find the probability of:

(a) Exactly 2 heads

(b) At least 4 heads.

Solution:

The repeated tossing of the coin is an example of a Bernoulli trial. According to the problem:

Number of trials: n=5

Probability of head: p= 1/2 and hence the probability of tail, (1-p) =1/2

(a) For exactly two heads:

x=2 (number of getting head/success)

P(x=2) = $n_{C_x} P^x (1-P)^{n-x} = \frac{5!}{2!(3)!} \times 0.5^2 \times (0.5)^3$

P(x=2) = 5/16

(b) For at least four heads,

P(x $\geq$ 4) = P(x = 4) + P(x=5)

Hence,

P(x = 4) = $\frac{5!}{4!(1)!} \times 0.5^4 \times (0.5)^1$= 5/32

P(x = 5) == 0.5^5 = 1/32

Therefore,

P(x $\geq$ 4) = 5/32 + 1/32 = 6/32 = 3/16

How the distribution looks like, we will take an example in Python…..

Geometric Distribution

- It occurs when you count the number of independent Bernoulli trials until the first success
- In this case, the number of trials *k* will be distributed with Geometric Distribution

In other words,

It is the probability that first success occurs after *k* number of trials. If p is the probability of success each trial,

then the probability that success occurs on the k^{th} trial is given by the formula

$$PMF(k) = (1-p)^{k-1}p$$

Example: a person is throwing dice and will stop once he gets 5. Calculate the probability that the person gets number 5 for the first time and second time.

Answer:

Since there are 6 possible outcomes, the probability of success p =1/6= 0.17. Therefore, the probability of failure 1 – p = 1 – 0.17 = 0.83

The person gets number 5 for the first-time k = 1. Substituting the values of k, p and 1-p in distribution, we have

$$PMF(k) = (1-p)^{k-1}p = (0.83)^0 \times 0.17 = 0.17$$

The person gets number 5 for the second time k = 2. Substituting the values of k, p and 1-p in distribution, we have

$$= (0.83)^1 \times 0.17 = 0.14$$

How the distribution looks like, we will take an example in Python…..

Probability Distribution for Continuous Random variables

Uniform Distribution

- It is a probability distribution that has constant value in each interval
- It is also known as Rectangular Distribution
- Used to generate random values

An example of uniform distribution

Random Number Generation: When generating random numbers within a specified range, a uniform distribution is often used. For example, if you need to generate random integers between 1 and 100, a uniform distribution ensures that each number in that range has an equal probability of being selected.

Exponential Distribution

- It provides a way to find the probabilities in time for a process
- Traditionally, it is used for modelling time to failure of electronic components
- This represents a process in which events occur continuously and independently at a constant average rate

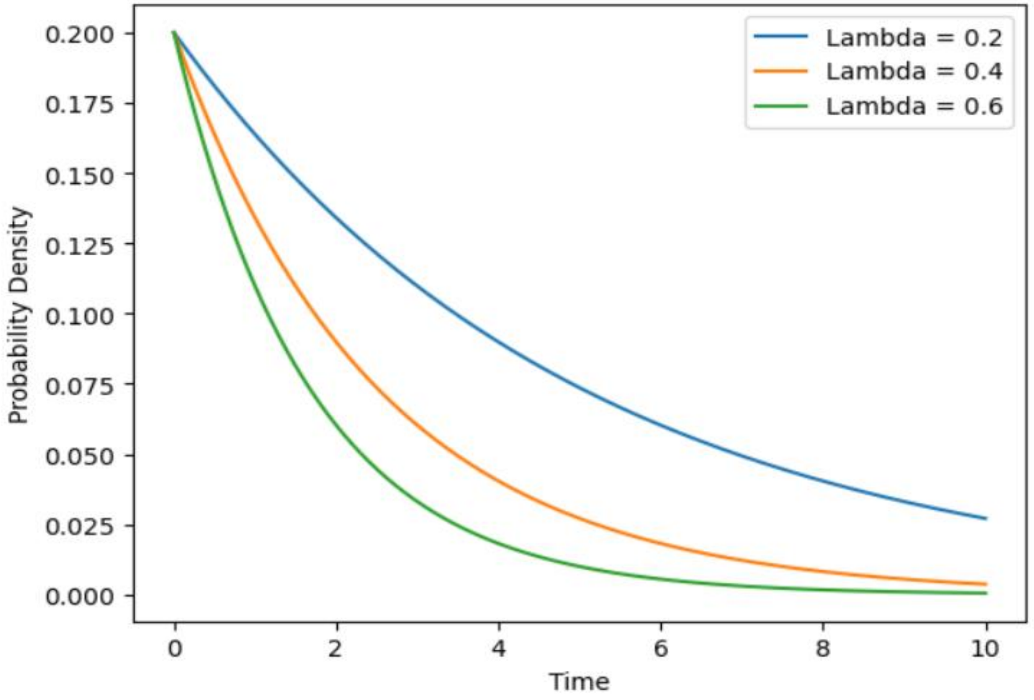
Exponential Distribution Formula

The continuous random variable x have an exponential distribution, if it has the following probability density function:

$$PDF(x) = \lambda e^{-\lambda x}$$

Where λ (lambda) is the mean rate parameter, which determines the average number of events per unit of time. x is the time variable

Exponential Distribution Cont.



Again, λ represents the mean rate or mean occurrence rate of events.
It is **the average number of events that occur in a specific unit of time**, such as the average number of customers arriving at a store per hour, or the average number of emails received per day.

λ is a positive real number, and it determines the shape of the exponential distribution. A higher value of λ indicates that events occur less frequently, while a lower value of λ indicates that events Occur more frequently.

λ is also related to the mean (μ) and variance (σ^2) of the exponential distribution. The mean is the average time between events, and it is equal to the reciprocal of lambda:

$$\mu = \frac{1}{\lambda}$$

The variance is the square of the mean, and it is also equal to the reciprocal of lambda squared:

$$\sigma^2 = \frac{1}{\lambda^2}$$

Example: Assume that you usually get 2 phone calls per hour. Calculate the probability, that a phone call will come within the next hour.

Solution:

It is given that, 2 phone calls per hour. So, it would be expected that one phone call every half-an-hour. So, we can take

λ = 2 (It is the number of calls per hour)

So, the computation is as follows:

$$PDF(0 \leq x \leq 1) = \int_0^1 \lambda e^{-\lambda x} = \int_0^1 2e^{-2x} = 0.86466$$

Remember: $\int e^{ax} dx = \frac{1}{a} e^{ax} + C$

Therefore, the probability of receiving the phone calls within the next hour is 0.393469

Normal Distribution

- One of the most important probability Distribution in the field of statistics
- It is also called a Gaussian distribution or Bell curve distribution.
- Fits several natural phenomena such as Measurement error, heights, IQ score, and Blood pressure and so on

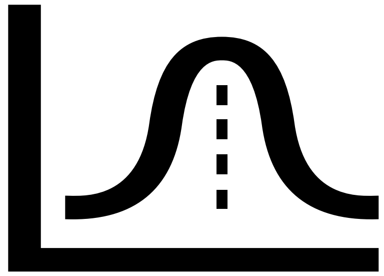
Normal Distribution Formula

The probability density function of normal or gaussian distribution is given by

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{\frac{-(x-\mu)^2}{2\sigma^2}}$$

Where, x is the variable, μ is the mean, and σ is the standard deviation

Example: In an exam, the marks are normally distributed with a mean of 65 and standard deviation of 9. Find the probability of the students marks that are



a) Less than 54
b) At least 80
c) Between 70 and 87

Answer:

a)

Using the PDF of the normal distribution, the probability density at a specific point x is given by the formula:

$$f(x,\mu,\sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{\frac{-(x-\mu)^2}{2\sigma^2}}$$

For this problem, x = 54, μ = 65, and σ = 9. Plugging these values into the formula, we get:

$$f(54) = \frac{1}{9\sqrt{2\pi}} e^{\frac{-121}{162}} = 0.021$$

Therefore, the probability that student get mark less than 56 is equal to the area under the curve to the left of 56, which is the cumulative probability. We can obtain this by integrating the PDF from negative infinity to 56:

$$P(\leq 54) = \int_{-\infty}^{54} f(x)\, dx$$

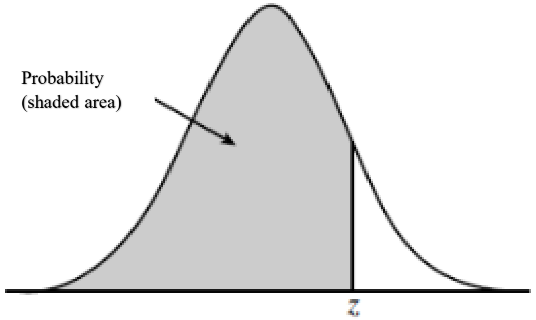
The answer should be 0.1112

An easy way to solve this problem is to use Z-score table

- zscore for the value 54 is

$$Z = \frac{x-\mu}{\sigma} = \frac{54-65}{9} = -1.2222$$

From Z-score table, we will look to the probability of z = -1.22



z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
−3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
−3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
−3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
−3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
−3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
−2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
−2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
−2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
−2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
−2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
−2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
−2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
−2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
−2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
−2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
−1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
−1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
−1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
−1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
−1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
−1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
−1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
−1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
−1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
−1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379

$$P(\leq 54) = P(Z = -1.22) = 0.1112$$

b) At least 80

- zscore for the value 80 is

$$Z = \frac{x-\mu}{\sigma} = \frac{80-65}{9} = 1.666$$

$$P(\geq 80) = 1 - P(\leq 80) = 1 - P(Z = 1.67) = 1 - \ 0.9525 = 0.0475$$

a) Between 70 and 86

- zscore for the value 70 is

$$Z = \frac{x-\mu}{\sigma} = \frac{70-65}{9} = \ 0.555$$

- zscore for the value 86 is

$$Z = \frac{x-\mu}{\sigma} = \frac{86-65}{9} = \ 2.333$$

$$P(70 \ \geq x \ \geq \ 86) = \ P(\leq 86) - \ P(\leq 70)$$

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964

$P(Z = 2.33) - \ P(Z = 0.55)$ = 0.9901-0.7088= 0.281

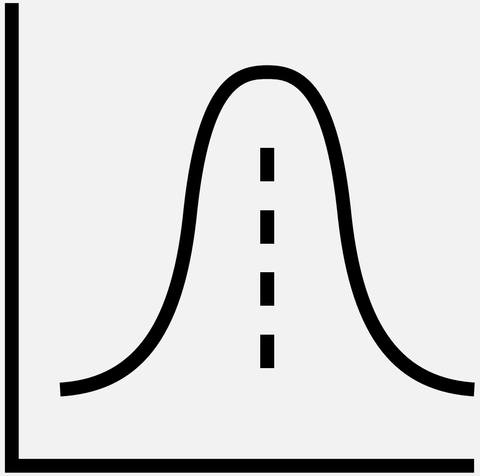
Let us demonstrate how to implement this in python …………..

Python

This material is prepared and edited by Dr. Khaled Mohamad

Transforming data to follow a normal distribution.

The normal distribution has several important properties that make it a valuable tool in statistical and data science analyses.



- It aligns with the Central Limit Theorem, which allows us to make inferences about populations based on smaller samples.
- Transforming data to approximate a normal distribution allows us to apply many statistical methods and hypothesis test techniques and draw valid conclusions from the data.
- Normal distribution transformation is often used to preprocess data in data science. It helps ensure that the data meets the assumptions of machine learning models, such as linearity and normality, leading to improved model performance.

Methods to transform a distribution to approximate a Normal Distribution

The following are the commonly used techniques:

Z-score Transformation: It transforms the original data point from its original distribution to a standard normal distribution with a mean of 0 and a standard deviation of 1. This transformation allows you to compare and analyze data points across different distributions, as they are all standardized to the same scale.

- The z-score transformation assumes that the original data follows a normal distribution. If the data is not normally distributed, applying z-score transformation may not be appropriate or may lead to misleading results.
- Z-score transformation does not change the shape of the data distribution. If the original data has a skewed or non-normal distribution, the transformed data will still exhibit the same distributional shape, but with a mean of 0 and a standard deviation of 1.

Logarithmic Transformation: Taking the logarithm of the data can be useful when **the distribution has a positive skewness (skewed to the right).** This transformation can help make the data more symmetric.

Square Root Transformation: Taking the square root of the data is **effective for reducing right skewness**. It can be particularly useful when the data contains many small values.

Box-Cox Transformation: The Box-Cox transformation is a power transformation that can handle a wide range of data distributions. It involves applying a parameter, lambda (λ), to the data to achieve the desired transformation. By varying the value of lambda, different transformations can be applied, including logarithmic, square root, and reciprocal transformations.

Power Transformation: Power transformations involve raising the data to a power other than 1. This can be **useful for distributions with heavy tails or skewness**. Common power transformation reciprocal transformation.

An important note: These transformations aim to approximate a normal distribution, but the effectiveness **may vary depending** on the specific data and its underlying characteristics.
Additionally, transformed data should be interpreted with caution, as the original scale and meaning may be altered.

Python

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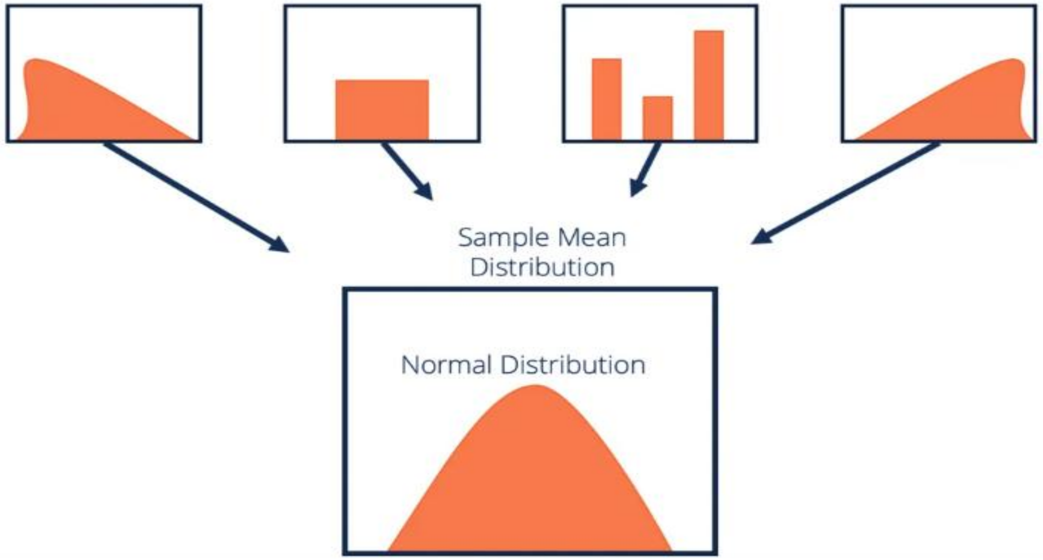
Central Limit Theorem

The central limit theorem (CLT) states that the distribution of sample means approximates a normal distribution as the sample size gets larger, regardless of the population's distribution.

The Central Limit Theorem (CLT) is a fundamental concept in statistics and probability theory. The key aspects of CLT are as follows:

- For a large enough sample size, the distribution of sample means will be approximately normally distributed, regardless of the shape of the population distribution.



- The average of these sample means will be equal to the population mean.

- The standard deviation of these sample means, called the **standard error** of the mean, will be equal to the population standard deviation divided by the square root of the sample size.

CLT cont.

- The CLT is valuable because it allows us to make inferences and draw conclusions about a population based on a sample, even if the population distribution is not known.

- It is commonly used in hypothesis testing, confidence interval estimation, and constructing sampling distributions.

- The applicability of the CLT relies on certain assumptions, such as the random sampling of observations and the independence of the sampled values.

- The Central Limit Theorem has wide-ranging applications in various fields, including finance, economics, biology, and social sciences.

A practical Example for CLT

Imagine you are conducting a survey in your school to estimate the average height of students. You collect data from a random sample of 50 students and calculate their heights. However, you cannot measure the height of every student in the school, so you want to use this sample to make an inference about the average height of all students.

According to the CLT, even if the heights of individual students are not normally distributed, the distribution of sample means (the average height of each sample) will tend to follow a normal distribution as the sample size increases.

To demonstrate this, you can perform the following steps:

- Collect the heights of the 50 students in your sample.
- Calculate the mean height of this sample.
- Repeat steps 1 and 2 multiple times (let's say 1000 times) to create a distribution of sample means.
- Plot a histogram of these sample means.

What you will observe is that the distribution of sample means will likely approximate a bell-shaped curve, even **if the individual heights are not normally distributed**. This is the essence of the Central Limit Theorem. **As you increase the number of samples or the sample size, the distribution of sample means will converge towards a normal distribution**.

This means that you can use this distribution to make inferences about the average height of all students in the school. You can calculate confidence intervals or perform hypothesis tests based on the assumption of normality, even if the original data did not follow a normal distribution.

Let us verify CLT using Python …………………………..

Python

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